

**Course Code: CSE-1223****Course Title: Discrete Mathematics****Credit Hours: 3****Contact Hours: 3 lecture hours per week****Type: Core, Engineering****Prerequisite: None**

Course Assessments	CIE: Continuous Internal Evaluation	Attendance	10 Marks
		Class test/ Assignment/ Quizzes	10 Marks
		Mid-term	30 Marks
	SEE: Semester End Examination		50 Marks

Course Objectives:

1. To obtain general knowledge about the areas of discrete mathematics and algebra
2. To understand a variety of methods used to construct mathematical proofs
3. To acquire an insight into applications such as coding and design

Course Learning Outcomes(CLOs) :

Upon successful completion of this course, students will be able to:

#	CLO Description	Weightage (%)
CLO1	Understand fundamental concept of different discrete structures like set, function, relation, graph, tree etc. and their properties	20
CLO2	Understand the concept of different types of formal logic and mathematical reasoning	20
CLO3	Apply the concept of formal logic and mathematical reasoning	40
CLO4	Apply various concepts of number theory and combinatorics	20

Mapping of CLO-PLO:

#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
CLO1	Understand fundamental concept of different discrete structures like set, function, relation, graph, tree etc. and their properties	PLO1	Cognitive/Understand	Lecture, Class discussion, Note	Exam, Quiz, Assignment
CLO2	Understand the concept of different types of formal logic and mathematical reasoning	PLO1	Cognitive/Understand	Lecture, Class discussion, Note	Exam, Quiz, Assignment



CLO3	Apply the concept of formal logic and mathematical reasoning	PLO1	Cognitive/Apply	Lecture, Class discussion, Note	Exam, Quiz, Assignment
CLO4	Apply various concepts of number theory and combinatorics	PLO1	Cognitive/Apply	Lecture, Class discussion, Note	Exam, Quiz, Assignment

8 Stanza Syllabus:

Stanza	Topics
1.	Set: Introduction to Set, Set operation, Representation of Sets, Algebraic Properties of Set, Computer representation of set.
2.	Logic: Introduction to Logic, Propositional Calculus, propositional equivalences, Predicate and quantifier, Translating sentence into logical expressions.
3.	Function & Relation: Introduction to function, Some important functions, Properties of function, Rate of Growth: Big O Notation, Sequence and summation. Introduction to Relation, Representation of Relation, Properties of Relation, Composition of Relation, Equivalence Relations, Partition, Closures of Relation.
4.	Number Theory: Theorem of Arithmetic, Modular Arithmetic, GCD, LCM, Prime Number, Congruence, Application of Congruence, Application of Number Theory, Chinese Remainder Theory.
5.	Mathematical reasoning: Methods of Proof, Mathematical Induction, Recursion.
6.	Counting: Basic Counting principle, Inclusion-Exclusion principle, Application of Sum rule and Product rule, Pigeon hole principle, Permutation and Combination, Binomial coefficients, Recurrence relations.
7.	Graph: Graph terminology, Representation of graph, Graph Isomorphism, Graph connectivity, Euler and Hamilton paths, Shortest Path Problems, Planner Graphs, Graph coloring.
8.	Tree: Definition of Trees, Application of trees, Tree Traversal, Trees and sorting, Spanning trees, Minimum spanning trees.

Text Books:

1. Kenneth H. Rosen Discrete Mathematics and Its Applications 7th edition McGraw-Hill 2012 ISBN-13: 978-0073383095 ISBN-10: 0073383090

Reference Books:

1. Seymour Lipschutz and Marc Laris Lipson Schaum's Outline of Theory and Problems of Discrete Mathematics 2nd edition McGraw-Hill 1997 ISBN-13: 978-0071615860 ISBN-10: 0071615865
2. J. P. Tremblay and R. P. Manohar Discrete Mathematical structures with applications to Computer Science McGraw-Hill 1975 ISBN: 0070651426
3. Donald F. Stanat and David F. McAllister "Discrete Mathematics in Computer Science " Prentice-Hall 1977 ASIN: B01A64QCMY



4. "B. Kolman R.C. Busby and S. Ross" Discrete Mathematical Structures 3rd edition
"Prentice- Hall " 1996 ISBN-13: 978-0-13-229751-6 ISBN-10: 0-13-229751-5
5. C. L. Liu Elements of Discrete Mathematics 2nd edition McGraw-Hill 1985 ISBN-10:
0070381305 ISBN-13: 978-0070381308